SOFTWARE REQUIREMENTS SPECIFICATION

Retail Sales Forecasting Dashboard

SQL + Power BI Project

Version	1.0
Date	March 2026
Dataset	Kaggle — Store Sales Forecasting
Database	MySQL Workbench 8.0
Visualization	Microsoft Power BI Desktop
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Status	Completed

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) documents the Retail Sales Forecasting Dashboard — a business intelligence project built using MySQL Workbench and Microsoft Power BI Desktop. It outlines what the project does, who it serves, what business questions it answers and how the data flows from a raw Kaggle dataset into an interactive dashboard with sales forecasting.

1.2 Project Overview

Retail businesses need to understand where their revenue is coming from, which products and regions are performing well, how discounts are affecting profitability and what sales might look like in the coming months. This project addresses all of those needs by transforming raw sales transaction data into clear, actionable visual insights.

The dashboard covers 4 years of retail sales data (2014–2017) across multiple US states, product categories and customer segments and projects future sales 12 months ahead using Power BI's built-in forecasting engine.

1.3 Scope

The project is designed to answer the following core business questions:

- How have total sales and profits trended over the past 4 years?
- Which product categories and sub-categories generate the most revenue?
- Which US states and regions are our strongest and weakest markets?
- Which customer segments — Consumer, Corporate or Home Office — contribute the most?
- Are discounts helping sales or hurting profitability?
- What can we expect sales to look like over the next 12 months?

To answer these questions the project covers:

- Dataset: 2,121 retail sales records from Kaggle spanning 2014 to 2017
- Database: MySQL Workbench 8.0 — all data cleaning and analysis done in SQL
- Analysis: 6 SQL queries producing KPIs, trends, category, region, product and discount insights
- Views: 5 clean SQL views created as ready-to-use data sources for Power BI
- Dashboard: 2-page Power BI report with 10 visuals, 4 KPI measures and a 12-month forecast

1.4 Definitions

Term	Definition
SRS	Software Requirements Specification — this document
SQL	Structured Query Language — used in MySQL for all data operations
Power BI	Microsoft business intelligence tool used for the dashboard and forecasting
KPI	Key Performance Indicator — a measurable value that shows business performance
CSV	Comma Separated Values — the raw file format of the Kaggle dataset
DAX	Data Analysis Expressions — Power BI formula language used to create measures
View	A saved SQL query in MySQL that acts like a virtual table
Profit Margin	Profit divided by Sales expressed as a percentage — shows how much of each sale is actual profit
Discount Band	A grouping of orders by discount level (0%, 1-10%, 11-20%, 21-30%, Over 30%)
Forecast	A statistical prediction of future sales values based on historical patterns

2. Stakeholders

The following stakeholders are involved in or affected by this project:

Stakeholder	Role	Interest in This Project
Sales Manager	Primary business user	Wants to understand which regions, products and segments are driving revenue and where to focus sales efforts
Finance Team	Business user	Needs visibility on profit margins and the financial impact of discounting strategies on overall profitability
Marketing Team	Business user	Wants to identify high-value customer segments and top performing product categories to target campaigns effectively
Operations Team	Business user	Interested in shipping days, order volumes by region and fulfilment patterns to optimize logistics
Business Analyst	Project developer	Responsible for building the data pipeline, writing SQL queries, creating views and designing the Power BI dashboard
Senior Management	Executive viewer	Uses the high-level KPIs and 12-month sales forecast to support strategic planning and investment decisions
Portfolio Reviewer	External evaluator	Reviews the project to assess data analytics skills, SQL proficiency and Power BI dashboard quality

3. Functional Requirements

Functional requirements describe what the system must do — the features and capabilities it must deliver to meet business needs.

3.1 Data Management Requirements

ID	Requirement	Business Justification
FR-01	The system must store all 2,121 sales records in a structured MySQL database	Centralised storage ensures all analysis is based on a single consistent source of truth
FR-02	The system must convert all date fields to a standard DATE format	Mixed date formats in the raw data would cause incorrect trend analysis and wrong shipping day calculations
FR-03	The system must calculate shipping days for every order	Shipping duration is a key operational metric — it helps identify fulfilment bottlenecks
FR-04	The system must flag and record data quality issues in the data_error column	Transparency about data quality issues allows business users to trust and audit the dashboard results
FR-05	The system must verify zero NULL values in all key business columns	NULL values in sales, profit or order fields would produce inaccurate KPI totals

3.2 Analysis Requirements

ID	Requirement	Business Justification
FR-06	The system must calculate total revenue, total profit, profit margin and total orders as KPIs	These four metrics are the minimum viable set for any retail performance review
FR-07	The system must show monthly sales and profit trends over the full 4-year period	Month-by-month trends reveal seasonality, growth patterns and sudden dips that annual totals hide
FR-08	The system must break down sales and profit by product category and sub-category	Category analysis tells the business which product lines to invest in and which to review
FR-09	The system must show performance by US state and region	Geographic analysis helps identify expansion opportunities and underperforming markets
FR-10	The system must measure the impact of discount levels on profit margins	Understanding the discount-profit relationship prevents the business from offering discounts that cause net losses
FR-11	The system must compare sales performance across Consumer, Corporate and Home Office segments	Segment analysis informs targeted sales and marketing strategies for each customer group

3.3 Dashboard Requirements

ID	Requirement	Business Justification
FR-12	The dashboard must display 4 KPI cards showing Total Sales, Total Profit, Profit Margin % and Total Orders	Executives need to see top-line numbers at a glance without navigating to detailed charts
FR-13	The dashboard must include a geographic map showing sales by US state	Visual maps are faster to read than tables when comparing performance across 40+ states
FR-14	The dashboard must include a 12-month sales forecast with a 95% confidence interval	The forecast gives management a data-driven basis for setting targets and planning inventory
FR-15	All visuals must use a consistent dark navy theme across both pages	A consistent theme improves readability and presents the dashboard as a professional deliverable

4. Non-Functional Requirements

Non-functional requirements describe how the system performs — qualities like usability, reliability and maintainability rather than specific features.

4.1 Usability

ID	Requirement	Detail
NFR-01	The dashboard must be readable by non-technical business users	All chart titles, axis labels and KPI card labels must use plain business language — no SQL column names visible to end users
NFR-02	Key insights must be visible without scrolling on a standard 1080p screen	The most important KPIs and charts must fit within a single screen view on each page
NFR-03	Chart colors must encode meaning clearly	Green bars indicate positive profit margins, red bars indicate losses — color is not decorative but informational

4.2 Reliability

ID	Requirement	Detail
NFR-04	All SQL queries must produce consistent results on repeated execution	Queries must be deterministic — running the same query twice must return the same output
NFR-05	The data cleaning process must result in zero NULL values in key columns	NULL values in sales or profit fields would silently undercount totals and mislead business decisions
NFR-06	Shipping days must always be zero or positive	Negative shipping days are logically impossible and indicate a data error that must be resolved before analysis

4.3 Maintainability

ID	Requirement	Detail
NFR-07	All SQL queries must be clearly commented and organised by phase	Comments help any future analyst understand the purpose of each query without needing additional documentation
NFR-08	SQL views must be used as the data layer between MySQL and Power BI	Views decouple the dashboard from the raw table — if the table structure changes only the view needs updating not every chart
NFR-09	The Power BI file must be saved and exportable as PDF	The dashboard must be shareable in a static format for stakeholders who do not have Power BI installed

4.4 Performance

ID	Requirement	Detail
NFR-10	All SQL analysis queries must complete within 5 seconds on the 2,121-row dataset	Fast query execution ensures the analyst can iterate quickly during the development phase
NFR-11	Power BI visuals must load within 3 seconds of opening the report	Slow loading dashboards reduce user trust and engagement during stakeholder presentations

5. Tools and Dataset

5.1 Tools Used

Tool	Version	Purpose
MySQL Workbench	8.0	Database creation, data import, cleaning, analysis and views
Power BI Desktop	Latest	Dashboard creation, DAX measures and forecasting visual
Kaggle	Web platform	Source of the retail sales CSV dataset

5.2 Dataset Details

Attribute	Details
Source	Kaggle — Store Sales Forecasting Dataset
File name	stores_sales_forecasting.csv
Total rows	2,121 records
Total columns	21 columns
Date range	2014 to 2017 (4 years of transactions)
Key columns	Order ID, Order Date, Ship Date, Customer, Segment, Region, Category, Sub-Category, Sales, Quantity, Discount, Profit

6. Database Schema

Database Name: data_store | Primary Table: orders

Column Name	Data Type	Description
row_id	BIGINT	Unique row identifier
order_id	TEXT	Unique order identifier (e.g. CA-2016-137043)
order_date	TEXT	Original order date text — mixed formats in source file
ship_date	TEXT	Original ship date text — mixed formats in source file
ship_mode	TEXT	Shipping method used (Standard Class, First Class, Second Class, Same Day)
customer_id	TEXT	Unique customer identifier
customer_name	TEXT	Full name of the customer
segment	TEXT	Customer segment (Consumer, Corporate, Home Office)
country	TEXT	Country of the order
city	TEXT	City of the order
state	TEXT	US state of the order
postal_code	VARCHAR(10)	Postal / ZIP code
region	TEXT	Geographic region (East, West, Central, South)
product_id	TEXT	Unique product identifier
category	TEXT	Product category (Furniture, Technology, Office Supplies)
sub_category	TEXT	Product sub-category (Chairs, Tables, Phones, Binders etc.)
product_name	TEXT	Full product name
sales	DECIMAL(10,4)	Order sales amount in USD
quantity	INT	Number of units ordered
discount	DECIMAL(5,4)	Discount rate applied to the order (0.0 = no discount, 0.8 = 80% discount)
profit	DECIMAL(10,4)	Profit or loss in USD — negative values indicate loss-making orders
order_date_new	DATE	Cleaned order date — converted from text to DATE format during Phase 2
ship_date_new	DATE	Cleaned ship date — converted from text to DATE format during Phase 2
shipping_days	INT	Calculated number of days between order date and ship date

Column Name	Data Type	Description
data_error	VARCHAR(50)	Records any data quality flag identified during the cleaning phase (e.g. missing date, invalid value)

7. Data Cleaning

All data cleaning is performed using SQL queries in MySQL Workbench after importing the Kaggle CSV file. The goal is to produce a reliable, analysis-ready dataset with no data quality issues that would affect business conclusions.

7.1 Cleaning Steps Performed

Cleaning Step	Issue Found	SQL Solution Applied
Date conversion	Mixed formats — some rows DD-MM-YYYY, others MM/DD/YYYY in the same column	CASE + STR_TO_DATE using LIKE '%-%' and LIKE '%/%' patterns to handle both formats
Shipping days	No shipping duration column existed in the raw dataset	ALTER TABLE to add shipping_days column + DATEDIFF(ship_date_new, order_date_new)
Whitespace trimming	Leading and trailing spaces found in key text columns	TRIM() applied to customer_name, city, state, region, category, sub_category, segment and ship_mode
NULL check	Verified all business-critical columns contain no NULL values	COUNT(*) - COUNT(col) pattern applied to all key columns — result: 0 nulls found
Duplicate check	2 duplicate order + product ID combinations identified	Retained in the dataset — confirmed as valid repeat line items within the same order
Negative profit review	714 orders found with negative profit totalling -$60,936	Retained — these are valid records caused by heavy discount rates applied to orders

7.2 Data Quality Results After Cleaning

Metric	Value	Result
Total rows in database	2,121	Pass
NULL values in key business columns	0	Pass
Negative shipping days	0	Pass
Average shipping days per order	41.3 days	Pass
Maximum shipping days recorded	214 days	Pass
Loss-making orders (negative profit)	714 orders — 33.7% of total	Valid — retained for analysis

8. SQL Analysis Queries

Six SQL queries are developed in MySQL Workbench to answer the core business questions identified in the project scope. Each query targets a specific area of the business.

#	Query Name	Business Question Answered	Key Output
8.1	Overall KPIs	What is the overall health of the business?	Revenue $742K, Profit $18K, Margin 2.49%, Orders 1,764
8.2	Monthly Sales Trend	How have sales changed month by month over 4 years?	48 monthly data points showing consistent year-over-year growth
8.3	Category Performance	Which product lines are most and least profitable?	Furniture leads revenue; Chairs is top sub-category
8.4	Regional Analysis	Where in the US are we strongest and weakest?	State and region breakdown identifying top and bottom markets
8.5	Top 10 Products	Which individual products generate the most revenue?	Ranked product list with sales, profit and units sold
8.6	Discount Impact	Are our discounts helping sales or hurting profit?	Discounts over 30% consistently result in net losses

9. SQL Views

Five views are created in MySQL using CREATE OR REPLACE VIEW. Rather than connecting Power BI directly to the raw orders table, these views provide clean pre-aggregated summaries — making the Power BI data model simpler, faster and easier to maintain.

View Name	Key Columns	Used In Power BI For
monthly_sales	year_month, month_start_date, total_sales, total_profit, total_orders, profit_margin_pct	Monthly trend line chart, all 5 KPI card measures, Forecast visual
category_performance	category, sub_category, total_sales, total_profit, units_sold, profit_margin_pct	Sales by Sub-Category horizontal bar chart
region_summary	region, state, city, total_sales, total_profit, total_orders	Total Sales by State bubble map visual
segment_summary	segment, total_sales, total_profit, total_customers, avg_order_value	Sales by Segment donut chart
discount_analysis	discount_band, num_orders, total_sales, total_profit, avg_margin_pct	Profit Margin by Discount Band column chart

10. Power BI Dashboard

10.1 Data Connection

The 5 SQL views are exported from MySQL Workbench as CSV files and loaded into Power BI Desktop using Home → Get Data → Text/CSV. Data types are corrected in Power Query Editor before loading — date columns set to Date type and numeric columns set to Decimal Number.

10.2 DAX Measures

Five DAX measures are created inside the monthly_sales table to power the KPI cards and derived calculations:

Measure Name	DAX Formula	Displayed On
Total Sales	SUM(monthly_sales[total_sales])	KPI Card
Total Profit	SUM(monthly_sales[total_profit])	KPI Card
Profit Margin %	DIVIDE(SUM([total_profit]), SUM([total_sales])) * 100	KPI Card
Total Orders	SUM(monthly_sales[total_orders])	KPI Card
Avg Order Value	DIVIDE([Total Sales], [Total Orders])	Calculated metric

10.3 Page 1 — Sales Overview

Visual Title	Chart Type	Data Source	Fields Used
Total Sales	KPI Card	monthly_sales	Measure: Total Sales
Total Profit	KPI Card	monthly_sales	Measure: Total Profit
Profit Margin %	KPI Card	monthly_sales	Measure: Profit Margin %
Total Orders	KPI Card	monthly_sales	Measure: Total Orders
Monthly Sales Trend	Line Chart	monthly_sales	X: year_month, Y: Total Sales
Sales by Sub-Category	Clustered Bar Chart	category_performance	Y: sub_category, X: total_sales, Legend: category
Sales by Segment	Donut Chart	segment_summary	Legend: segment, Values: total_sales

10.4 Page 2 — Insights & Forecast

Visual Title	Chart Type	Data Source	Fields Used
Total Sales by State	Bubble Map	region_summary	Location: state, Size: total_sales, Tooltip: total_profit
Profit Margin by Discount Band	Clustered Column Chart	discount_analysis	X: discount_band, Y: avg_margin_pct
Sales Forecast — Next 12 Months	Line Chart + Forecast	monthly_sales	X: month_start_date, Y: Total Sales + 12 month forecast at 95% confidence

10.5 Why Exponential Smoothing for Forecasting

Power BI uses the Exponential Smoothing algorithm (also known as ETS — Error, Trend, Seasonality) to generate the sales forecast. This algorithm was chosen for the following reasons:

Reason	Explanation
Handles seasonality well	Retail sales naturally peak and dip at certain times of year. Exponential smoothing detects and models these repeating seasonal patterns automatically when seasonality is set to 12 months
Gives more weight to recent data	Unlike a simple average that treats all historical months equally, exponential smoothing gives higher importance to more recent months. This means the forecast responds to recent business trends rather than being diluted by old data
Suitable for small datasets	With only 48 monthly data points (2014-2017) the dataset is too small for complex machine learning models. Exponential smoothing is statistically reliable on small time series and does not overfit
Built into Power BI natively	No external tools or scripts are required. Power BI applies the algorithm directly in the Analytics pane on a line chart making it accessible to business analysts without programming skills
Provides confidence intervals	The algorithm produces upper and lower confidence bands (set at 95%) alongside the forecast line. This gives business users a realistic range of outcomes rather than a single fixed prediction
Proven in retail forecasting	Exponential smoothing is a widely used and validated approach in retail demand forecasting and sales planning — making the results credible to business stakeholders

10.6 Forecast Configuration

Setting	Value	Reason
Forecast length	12 months	Projects one full year of future sales for annual planning
Confidence interval	95%	Industry standard confidence level for business forecasts
Seasonality	12 points	Captures annual seasonal patterns repeating every 12 months
Algorithm	Exponential smoothing (ETS)	Best fit for small retail time series with seasonality
Training data	2014 to 2017 (48 months)	Full 4-year history used to train the forecast model

10.7 Dashboard Theme

Element	Style	Color / Value
Canvas background	White	#FFFFFF
Visual background	Dusty Grape	#4a4e69
Visual border	Accent light blue	#00B4D8
Title font	White, Bold, Arial 14	Applied to all visual titles
Positive profit bars	Green	0% and 1-10% discount bands
Negative profit bars	Red	21-30% and Over 30% discount bands
Forecast line	Black dotted line	Power BI auto-generated forecast projection
Confidence band	Grey shading	Upper and lower bounds of the 95% confidence interval

11. Key Business Findings

11.1 Overall Performance

- Total revenue of $742K generated across 1,764 orders from 2014 to 2017
- Consistent year-over-year revenue growth from $157K in 2014 to $215K in 2017
- Overall profit margin of 2.49% — typical of competitive retail markets with heavy discounting

11.2 Category Insights

- Furniture is the highest revenue category with Chairs as the single top sub-category
- Tables ranks second, followed by Bookcases and Furnishings

11.3 Customer Segments

- Consumer segment generates the largest share of total sales across all 3 segments
- Corporate segment shows higher average order values compared to Consumer
- Home Office is the smallest segment by both revenue and order count

11.4 Discount Impact

- Orders with zero discount generate the highest profit margins across all discount bands
- Discounts above 20% consistently result in negative profit margins
- 714 orders resulted in a combined loss of $60,936 — all attributable to heavy discounting
- Discounts over 30% are the single largest driver of profitability loss in the business

11.5 Sales Forecast

- 12-month forecast for 2018 shows continuation of the upward growth trend seen 2014-2017
- Seasonal fluctuations are visible in both the historical data and the forecast projection
- 95% confidence interval provides management with a realistic best-case and worst-case range

12. Project Completion Summary

Phase	Name	Deliverable	Tool	Status
1	Data Import	2,121 rows loaded into MySQL orders table from Kaggle CSV	MySQL	Done
2	Data Cleaning	Dates converted, shipping days calculated, NULLs verified, data_error column added	MySQL	Done
3	Analysis Queries	6 SQL queries answering all core business questions	MySQL	Done
4	SQL Views	5 pre-aggregated views created as Power BI data sources	MySQL	Done
5	Dashboard	2-page Power BI report with 10 visuals and 5 DAX measures	Power BI	Done
6	Forecast	12-month sales forecast using exponential smoothing at 95% confidence	Power BI	Done

This project demonstrates a complete end-to-end business intelligence workflow using MySQL Workbench and Microsoft Power BI Desktop with a real-world Kaggle retail dataset. The dashboard is designed to answer practical business questions and is suitable for portfolio presentation, stakeholder demonstrations and interview discussions.